

Divit Rawal

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Research Interests: statistical/machine learning theory; generalization and implicit bias in overparameterized models; geometric statistics and information geometry; stochastic processes and statistical physics of learning.

EDUCATION

University of California, Berkeley

Berkeley, CA

B.A. Statistics, Physics, Computer Science

2023 – 2027

Relevant Coursework: Statistical Learning Theory[†]; Theoretical/Mathematical Statistics[†]; Convex Optimization; Learning for Dynamics and Control[†]; Probability and Random Processes; Statistical/Thermal Physics; Randomized Linear Algebra, Optimization, Large-Scale Learning[†]; Statistical Computing. ([†] denotes graduate-level coursework)

Teaching: Head TA (Sp26), Reader (Sp25) for Communication Networks (EE 122). Developed lab curriculum, wrote and graded exams, and led weekly recitation sessions.

PUBLICATIONS AND PREPRINTS

A Theory of Saddle Escape in Deep Nonlinear Networks

2026

Divit Rawal, Michael R. DeWeese

Under review, NeurIPS.

[\[arXiv\]](#)

Rao-Blackwellized Score Matching on Manifolds

2026

Divit Rawal

ICML, Structured Probabilistic Inference and Generative Modeling Workshop.

[\[PDF\]](#)

Minimax Rates for Hyperbolic Hierarchical Learning

2026

Divit Rawal, Sriram Vishwanath

Preprint.

[\[arXiv\]](#)

ALPHANSO: Open-Source (α, n) Neutron Source Terms

2026

Divit Rawal, Anthony J. Nelson, William Zywiec, Daniel Siefman

2026 American Nuclear Society Student Conference.

[\[arXiv\]](#)

2026 Institute of Nuclear Materials Management Annual Meeting.

[\[GitHub\]](#)

Under review, Nuclear Instruments and Methods in Physics Research Section A.

AWARDS AND GRANTS

Best Paper Award (ANS Student Conference 2026)

Apr 2026

- Best Paper in Mathematics, Computation, and AI Applications at the American Nuclear Society Student Conference for “ALPHANSO: Open-Source Modeling of (α, n) Source Terms,” selected from ~ 250 submissions.

VESSL AI Academia Grant

Aug 2025

- Awarded academic grant for independent research into embeddings and information geometry; sole undergraduate awardee among recipients primarily comprising Ph.D. students and postdoctoral researchers.

RESEARCH EXPERIENCE

Lawrence Livermore National Laboratory

Mar 2025 – Present

Researcher (Nuclear Science and Security Consortium Fellow)

Livermore, CA

- Developed [ALPHANSO](#), an open-source computational framework for deterministic modeling of (α, n) neutron production using continuous slowing-down approximations and modern evaluated nuclear data.
- Formulated and implemented stochastic and deterministic transport models for neutron yield and spectral prediction, benchmarking against legacy codes and experimental datasets.
- Conducted quantitative validation against experimental neutron yield and spectral benchmarks, analyzing model error and sensitivity to nuclear cross-section data; demonstrated improved predictive accuracy and extensibility relative to legacy SOURCES-based pipelines.

Berkeley Artificial Intelligence Research (BAIR)

Sep 2024 – Present

Researcher, DeWeese Lab

Berkeley, CA

- Authored paper deriving an exact layer-imbalance identity for deep nonlinear networks and a critical-depth escape law, with matching theory and simulation under both symmetric and He-normal initialization.
- Analyzed in-context learning in kernel ridge regression via eigenlearning, deriving mode-wise error dynamics and conditions for generalization without representation change.
- Connected ICL behavior to spectral bias in kernel methods (eigenvalue decay, target alignment), clarifying how contextual examples induce effective adaptation in fixed-feature models.

UC Irvine, Department of Physics & Astronomy

Feb 2022 – Jul 2023

Researcher, Whiteson Lab

Irvine, CA

- Developed deep learning models using TensorFlow/Keras for high-energy particle collision analysis, achieving 90% accuracy (vs. 80% baseline) in particle identification.
- Simulated particle collisions using MadGraph, Pythia8, Delphes, and ROOT; implemented reconstruction algorithms in C++ and Python achieving 2% mass prediction error.
- Contributed to [arXiv:2412.01600](#) and [arXiv:2412.01548](#), both published in *Reports on Progress in Physics*.

INDUSTRY EXPERIENCE

Cisco Systems (Foundation AI)

Jun 2025 – Aug 2025

Software Engineer (Intern)

San Francisco, CA

- Contributed to post-training and evaluation of cybersecurity-specialized LLMs built on Llama 3.1 8B, including reasoning, instruction tuning, and preference alignment (RLHF) for security dialogue and workflows.
- Supported open-weight releases [Foundation-Sec-8B](#) and [Foundation-Sec-8B-Instruct](#), and contributed to the accompanying technical reports documenting data curation, training methodology, and benchmark evaluation.

ExperienceFlow AI

May 2024 – Sep 2024

Machine Learning Engineer (Intern)

Remote

- Modeled finite state machine (FSM) transition dynamics using transformer, state-space, and graph architectures, evaluating generalization and sample efficiency in low-label sequence prediction settings.
- Developed reinforcement-learning-based control policies (Deep Q-Learning, SARSA) for dynamic state reasoning systems, improving stability and convergence in sequential decision tasks.

Amazon

Aug 2023 – Dec 2023

Software Engineer (Intern)

Remote

- Implemented a K-means clustering module in Java for the [OpenSearch ml-commons](#) repository, contributing to distributed ML functionality and increasing unit test coverage from 66% to 78%.
- Diagnosed and resolved production data pipeline failures affecting 1M+ users, improving system reliability and reducing query latency in large-scale ML search infrastructure.

INVITED TALKS

Lawrence Berkeley National Laboratory

Mar 2026

ALPHANSO: An Open-Source Framework for (α ,n) Reactions.

Lawrence Livermore National Laboratory

Mar 2026

ALPHANSO: High-Fidelity (α ,n) Neutron Source Modeling.

TECHNICAL SKILLS

Programming: R, Python, C++, Java, MATLAB, Julia, SQL, Mathematica, $\LaTeX$.

Scientific Computing: NumPy, SciPy, Monte Carlo methods, randomized algorithms.

ML/AI: PyTorch, JAX, TensorFlow.